

Physics Assignment – Quantum Computing

Total 10 Marks

Q1. What is Quantum Computing? Explain its basic principles, key characteristics, and applications. [3 marks]

Q2. Explain the **Deutsch-Jozsa Algorithm**. Describe its objective, working principle, and significance in quantum computing. [2 marks]

Q3. Explain **Schrödinger's Cat Theory** and the **Smiley Theory** in the context of quantum computing and quantum concepts. Provide suitable examples to support your explanation. [3 marks]

Q4. What is **BB84**? Explain the BB84 Quantum Key Distribution protocol and its role in **Quantum Secret Messaging**. Describe its working steps and security aspects. [2 marks]